Bioinformatics

Lecture 4

Biological Sequence Alignment

- In Bioinformatics, a **sequence alignment** is a way of arranging the sequences of DNA, RNA, or protein to identify regions of similarity that may be a consequence of functional, structural, or evolutionary relationships between the sequences.

- It is an important first step toward structural and functional analysis of newly determined sequences.

- As new biological sequences are being generated at exponential rate, sequence comparison is becoming increasingly important to draw functional and evolutionary inference.

Biological Sequence Alignment

- The sequence alignment is made between a known sequence and unknown sequence or between two unknown sequences.

- The known sequence is called reference sequence.

- The unknown sequence is called query sequence.

Identity v/s Similarity

- When we talk about the comparison of two sequences than question arise that how we can compare the biological sequences and after comparison what will be the degree of comparison.
- There are two concepts for sequence analysis
- Identity
- Similarity
- Identity means the counting number of nucleotides or amino acids which exactly match when two biological sequences are matched.

Identity v/s Similarity

- For Example:
- Sequence 1: CATGCTT
- Sequence 2: CATGC
- Number of match = 5
- Smaller length = 5
- Sequence (1) = 7
- Sequence (2) = 5
- Formula for Identity:
- Identity = No. of Matches / smaller length × 100 = 100%

Identity v/s Similarity

- Similarity means the comparison between two different sequences calculated by alignment approach.

Introduction to Sequence Alignment

- **Sequence Alignment** is the process of lining up two sequences to achieve maximal levels of similarity for the purpose of assessing the degree of similarity and the possibility of homology.
- Aligned sequences of nucleotide or amino acid residues are typically represented as rows within a matrix. Gaps (-) are inserted between the residues so that identical or similar characters are aligned in successive columns.
- Computational approaches to sequence alignment generally fall into two categories: ***global*** *alignments* and ***local*** *alignments*.

Introduction to Alignment Approaches

- When we align the sequence that may be vary due to insertion and deletion of nucleotides and to calculate the similarity we need to align the sequence first. And there are two different approaches to align the sequence.

- Global Alignment

- Local Alignment

Local Alignment

- In Local alignment we compare one whole sequence with the one portion of other.

```
5' ACTACTAGATTACTTACGGATCAGGTACTTTAGAGGCTTGCAACCA 3'
             |||| |||||| |||||||||||||||
          5' TACTCACGGATGAGGTACTTTAGAGGC 3'
```

Global Alignment

- While in Global alignment we compare both sequence from end to end completely.

```
5' ACTACTAGATTACTTACGGATCAGGTACTTTAGAGGCTTGCAACCA 3'
   |||||||||||    |||||||  |||||||||||||| |||||||
5' ACTACTAGATT----ACGGATC--GTACTTTAGAGGCTAGCAACCA 3'
```

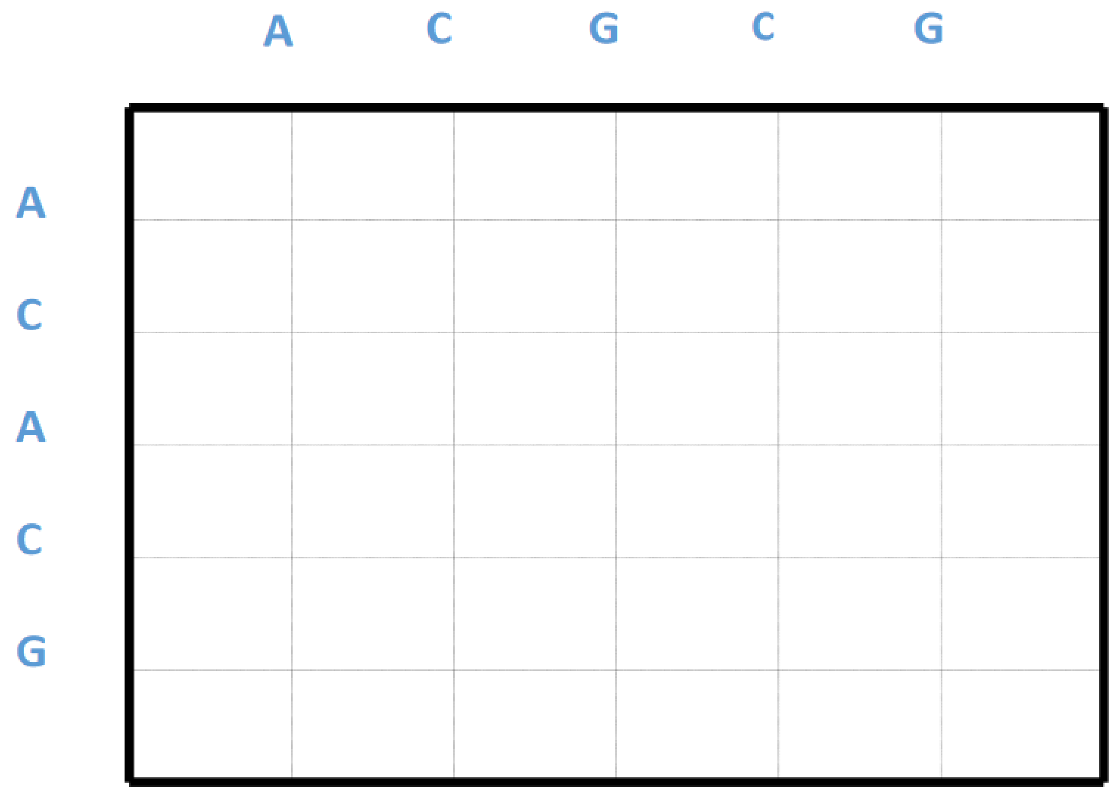
- **Local** alignment just focus on highly matching portions of sequence while in **Global** one whole sequence is compared with other one.

Local v/s Global

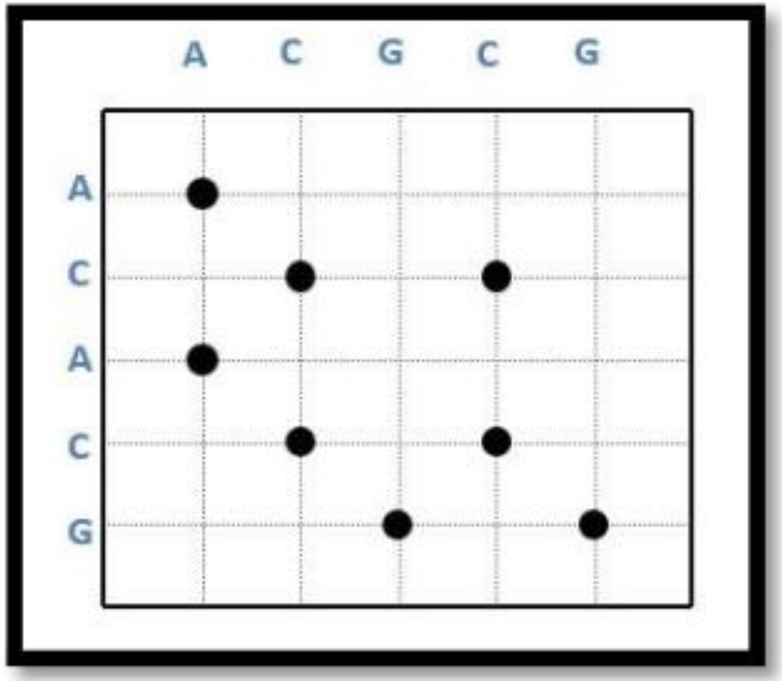
- When there is **Global** alignment which compare the whole sequence from end to end than why **local** alignment is done question arise.
- Because Local alignment have power to detect the smaller regions with high similarity and such matches are motifs or domains which remain hidden in case of protein function.
- Advantages: We can compare the different length sequences
- Conserved motifs can be determined from proteins
- Common function features can be identified.
- Local alignment is used to compare the segments for high matching sequencing.

Sequence Alignments Using Dot Plots

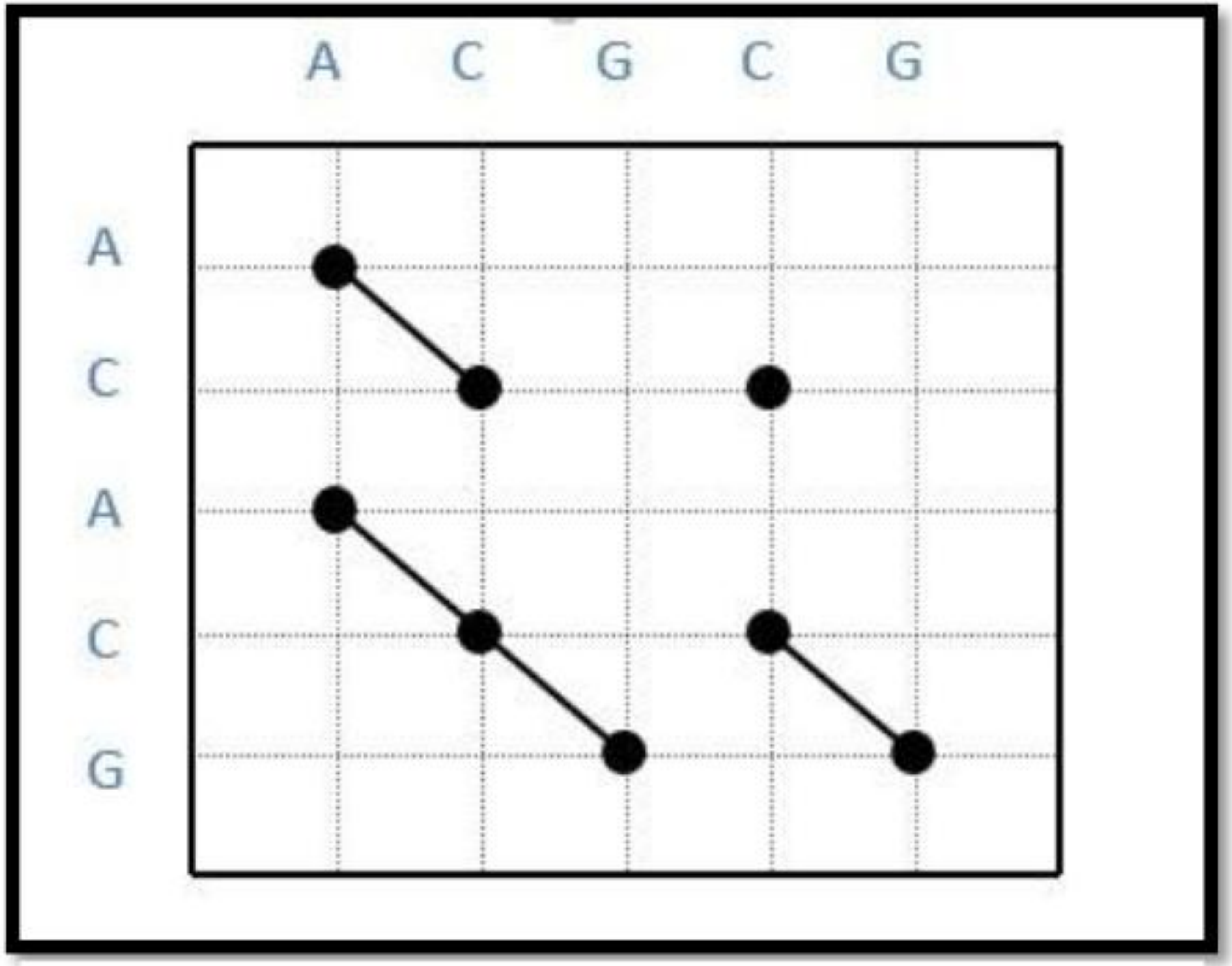
- To visualize the sequence alignment we have a method called Dot Plots. In this method the sequence is written top and left side of dot matrix grid.



Where one nucleotide or amino acid match with each other the dot is placed in grid position in each row for one time.

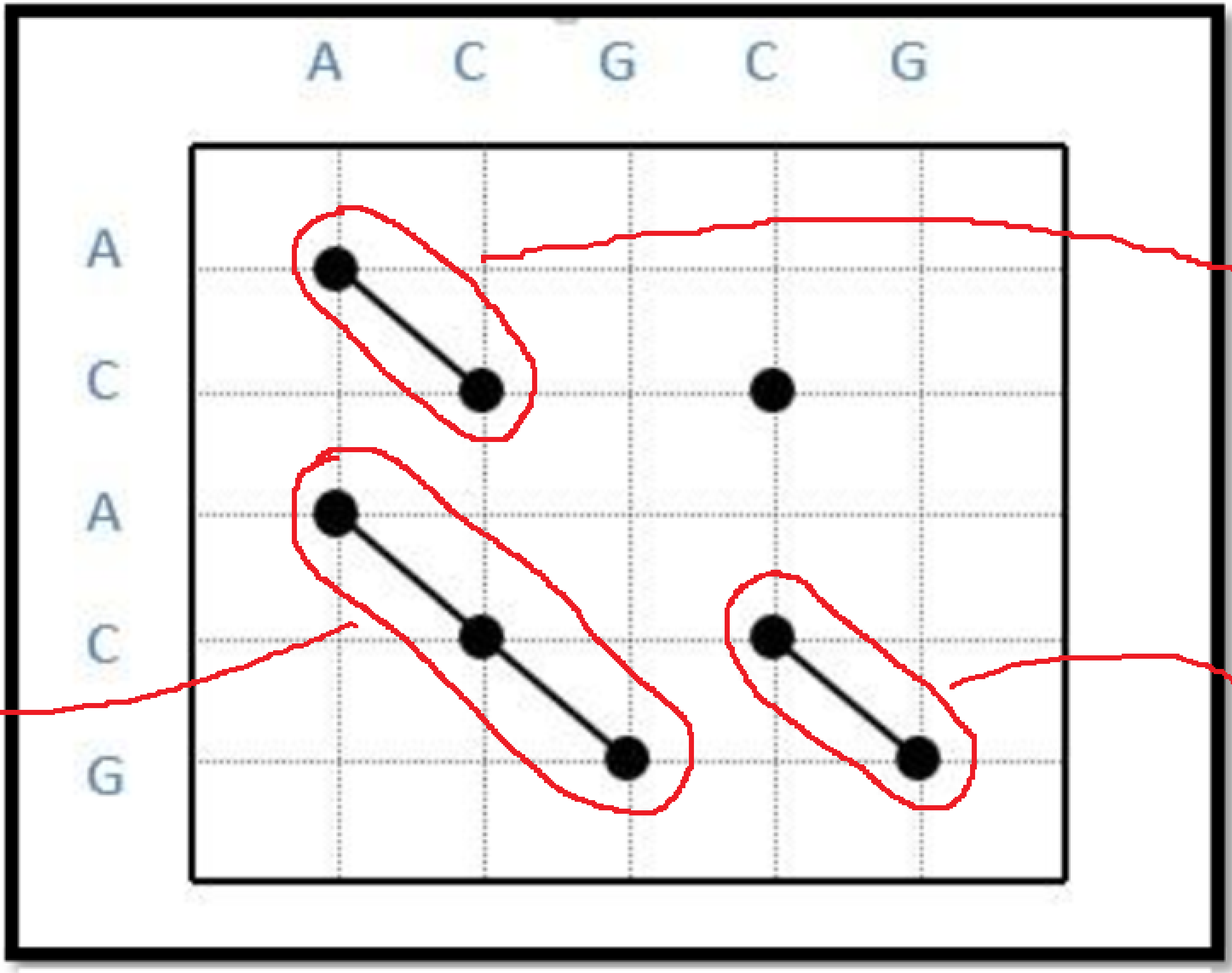


Similar dots are match with diagonal pattern and which remain separate differ from similar sequence



Dots on diagonal repeats the alignments and separate one give difference to the sequence

Match: +1
Mismatch:-1
Gap(-):-1




```
ACGCG...
||
AC...ACG
```

Score= -6

```
...ACGCG
     ||
ACA...CG
```

Score= -6

```
..ACGCG
  |||
ACACG..
```

Score= -1

Best Score

Dots on diagonal repeats the alignments and separate one give difference to the sequence

Benefits of DOT Plots

- Dot plots provides us the Global similarity between the two sequences and helps us to visualize the alignments of sequences and sequence repeats appear as diagonal stacks in plot.
- **Conclusion:**
- Dot plot help us to find the threshold difference among two sequences.